

Md. Navidul Hoque

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Career Objective

To build a long term career in technology driven and business oriented roles where I can apply my technical, analytical and problem solving skills to understand business operations, improve processes and support informed decision making while developing broader business and leadership capabilities.

Professional Summary

CSE graduate with 1+ years of backend engineering experience transitioning toward data analytics, business intelligence and technology driven business problem solving. Currently pursuing a Post Graduate Diploma in Data Science with Machine Learning and Artificial Intelligence. My technical background helps me understand business systems and data, while my analytical skills help me analyze business problems, identify patterns and support informed decisions.

Core Skills

- **Data Analytics:** Python, SQL, Pandas, Numpy, Statistics, Data Analysis, Power BI, Matplotlib, Excel
- **Machine Learning (Foundational Knowledge):** Machine Learning Fundamentals, Artificial Intelligence Fundamentals, Scikit-learn
- **Engineering Background (Production Experience):** TypeScript, JavaScript, Node.js, NestJS, PostgreSQL, REST APIs, Docker
- **Professional:** Problem Solving, Analytical Thinking, Communication, Collaboration, Adaptability, Business Oriented Thinking, Continuous Learning

Data Analytics Project

Bangladesh Healthcare Appointment Analytics

Stack: Python, Pandas, Matplotlib, Power BI, Excel, Scikit-learn, Machine Learning, Statistics, Data Analysis

[Github Repository](#) | [Descriptive Analysis Report](#) | [Predictive ML Analysis](#)

Descriptive Analytics:

- Analyzed 500 healthcare appointment records across 8 divisions of Bangladesh to identify no-show patterns, specialty demand and patient waiting time trends.
- Developed a reusable Python filtering function enabling division and specialty level analysis for interactive exploration.
- Identified Dhaka as having the highest no-show rate (**14.6%**) and General Physicians as having the longest average wait time (**11.6 days**).
- Conducted a **2023–2024** operational performance review in Excel, analyzing waiting time and no-show changes across divisions, age groups and specialties.
- Translated findings into **business implications and investigation areas** for improving appointment scheduling, capacity planning and patient attendance.
- Built analytical dashboards in **Matplotlib, Power BI and Excel** to communicate key trends and operational patterns.

Predictive Modeling:

- Built a machine learning pipeline in scikit-learn (Logistic Regression, Decision Tree, Random Forest) to predict which patients are likely to miss appointments, following proper data-splitting practices to ensure the results would hold up in real-world use.

- Created new features from the appointment date (day of week, quarter, weekday vs. weekend) and used stratified sampling to fairly handle the fact that only 11% of appointments were no-shows.
- Compared all three models using ROC-AUC and used cross-validation to tune their settings, selecting the Random Forest configuration with the most consistent validation performance for final risk scoring.
- Turned the model's predictions into simple risk levels (Low, Moderate, High, Very High) so the results could actually be used by a scheduling team.

Experience

Junior Software Engineer, TulipTech Ltd

Dhaka, Bangladesh | Nov 2025 – May 2026

- Built and maintained backend APIs for multiple internal products using NestJS and TypeScript.
- Designed PostgreSQL schemas and optimized SQL queries to support reliable, high performance access to production data.
- Used Supabase Realtime to handle live data updates across the system.
- Contributed to shared validation logic using Zod to keep data consistent across different services.

Backend Developer, Chetona

Dhaka, Bangladesh | Dec 2024 – Oct 2025

- Built REST APIs using Express.js to handle data requests across different user roles.
- Designed and structured MongoDB database schemas to keep data organized and queries efficient.
- Refactored the backend codebase using clean architecture to make it easier to maintain and scale.

Education

Post Graduate Diploma in Data Science with Machine Learning & Artificial Intelligence — PeopleNTech Institute of IT, 2026–2027

B.Sc. in Computer Science & Engineering — Military Institute of Science and Technology (MIST), 2020–2024

HSC — Dhaka Residential Model College, 2017–2019

SSC — Dhaka Residential Model College, 2015–2017

Certificates

Unleashing Cognitive Empowerment, Problem Solving & Decision Making

Passport to Earning Bangladesh | May 2026

[Certificate Link](#)

Workplace Communication Essentials, Communication Skills

Passport to Earning Bangladesh | May 2026

[Certificate Link](#)

Art of Problem Definition, Problem Definition

Passport to Earning Bangladesh | May 2026

[Certificate Link](#)